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Chemical Recycling, Kinetics, and Thermodynamics of Alkaline Depolymerization of Waste Poly(Ethylene Terephthalate) (PET)

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ABSTRACT

Depolymerization of PET in aqueous sodium hydroxide solution was undertaken in a batch process at 90–150°C and 1 atm by varying PET particle size in the range of 50–512.5 μm. Reaction time was also varied from 10–110 min to explore effect of particle size of PET and reaction time on batch reactor performance. Particle size of PET and reaction time required were optimized. Disodium terephthalate (TPA salt) and ethylene glycol (EG) remain in liquid phase. EG was recovered by salting-out technique. Disodium terephthalate was separated by acidification to obtain solid terephthalic acid (TPA). Produced TPA and EG were analyzed qualitatively and quantitatively. Yields of TPA and EG

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were almost equal to PET conversion. Depolymerization reaction rate was first order to PET concentration as well as first order to sodium hydroxide concentration. Acid value of TPA changes with reaction time. This indicates that PET molecule gets fragmented and hydrolyzes simultaneously with aqueous sodium hydroxide to produce EG and disodium terephthalate. Thermodynamics was also undertaken by determination of activation energy, Arrhenius constant, equilibrium constant, Gibbs free energy, enthalpy and entropy. Dependence of hydrolysis rate constant on reaction temperature was correlated by Arrhenius plot, which shows activation energy of 26.3 kJ/mol and Arrhenius constant of 427.2 L/min/cm^2.

Key Words: Poly(ethylene terephthalate); Depolymerization; Recovery; Kinetics; Acid value; Thermodynamics.

INTRODUCTION

Among different methods of polyester recycling, much attention has recently been paid to chemical recycling (Ben-Zu et al., 2001; Chen et al., 1991; Jun-ichi et al., 2000; Oku et al., 1997; Paszun and Spychaj, 1997; Yoshioka et al., 1994). Application of these methods depends on end use of recovered products. There have been several patented processes for depolymerization of PET (Deorr, 1986; Kirby et al., 1965; Schwartz, 1995; Tindall and Perry, 1991; Tindall et al., 1994). In these processes, depolymerization reactions were carried out under various conditions, and TPA was recovered (Adschiri et al., 1997; Campanelli et al., 1994; Cudmore, 1986; Kao et al., 1998a,b; Pusztaszeri, 1982; Rosen, 1992). Yields (%) of monomers are almost equal to PET conversion (Mishra and Goje, 2003; Mishra et al., 2003a–c). Moreover, processes of alkali hydrolysis of PET in the organic solvents have been reported by various researchers (Benzaria, 1993; Sako et al., 1997). Nevertheless, alkali hydrolysis was found to possess high activity for depolymerization of PET, very few reaction kinetic studies have been conducted, and they have not undertaken complete recovery of EG. Few researchers (Ben-Zu et al., 2001; Saw-Ming et al., 1996; Voss, 1989) have conducted kinetic studies but the rates observed were very low. In past there were no detailed data available about product characterization and kinetics, which were necessary to strengthen the explanation of alkali depolymerization of PET. Hydrolysis of PET is clearly a heterogeneous reaction at temperatures below its melting point (ranging from 250 to 264°C) (Yoshioka et al., 1998, 2001). Such considerations for a heterogeneous reaction have not been discussed in detail so far.

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Researchers (Ben-Zu et al., 2001) have conducted kinetic studies of depolymerization of PET in potassium hydroxide solution but the rates observed were very low. Results of their studies did not reach requirements of industrial depolymerization process. Hence, we have studied depolymerization of PET by using similar trends of researchers (Ben-Zu et al., 2001) during experiments and kinetic modelling. The major differences between the two studies, and the originality is given below:

Instead of using KOH, we have used NaOH because NaOH is cheaper than KOH. But in the work of researchers (Ben-Zu et al., 2001) kinetic parameter (k, reaction rate constant) calculated was very low that the value could not have any industrial significance. Moreover, in their (Ben-Zu et al., 2001) work the value of another kinetic parameter (Ea, activation energy) was high. Researchers (Ben-Zu et al., 2001) has not reported the comparison of activation energy (Ea) and Arrhenius constant (A) with the works of other researchers that is required to estimate the energy requirement for the reaction to decide the economics of the process during plant design of the process. By introducing cyclohexylamine (CHA) during reactions the required conversion could be obtained within the lesser time, temperature and pressure than that of the work of researchers (Ben-Zu et al., 2001). Researchers (Ben-Zu et al., 2001) report the depolymerization reactions of PET in a closed batch reactor at more than atmospheric pressure that requires costly reactor, their control and maintenance. A glass made batch reactor is cheaper and simple for use and problem of maintenance is not arising. As cost of glass reactor is low and easy for operation, so, glass reactor is most economical. Researchers (Ben-Zu et al., 2001) have not carried out PET wt (%) loss by varying the PET particle size that is required to understand the effect of PET particle on PET wt (%) loss. Effect of PET particle size and reaction time on acid value of TPA was not carried out in their (Ben-Zu et al., 2001) work that are required to develop a depolymerization scheme of PET during hydrolysis. Effects of PET concentration as a function of reaction time on PET weight (%) loss are not reported by researchers (Ben-Zu et al., 2001). This is required to take decision about the order of reaction with respect to PET concentration. Researchers (Ben-Zu et al., 2001) have not reported compensation effect that is required while designing the various process equipments. Researchers (Ben-Zu et al., 2001) have not investigated various thermodynamic parameters like equilibrium constant, Gibbs free energy, enthalpy and entropy (i.e. Ke, ΔG, ΔH and ΔS) that is required while transferring laboratory data in to pilot plant for commercialisation during design of various process equipments. Researchers (Ben-Zu et al., 2001) reports analysis of ethylene glycol (EG) without complete separation of EG from resulting liquid phase. Researchers (Ben-Zu et al., 2001) do not show complete separation of EG. Furthermore, during precipitation of TPA from

TPA salt researchers (Ben-Zu et al., 2001) reported the use of H_2SO_4. But for this purpose stoichiometric amount of HCl could be used because HCl is cheaper than H_2SO_4. HCl is comparatively less hazardous than H_2SO_4. In the use of H_2SO_4 there is possibility of carbonisation of TPA monomer. Hence, due to these reasons use of HCl is significantly better than H_2SO_4. Hence, there were various difficulties in the work of researchers (Ben-Zu et al., 2001) to transfer the process technology for commercialisation purpose. The published data depend on the specific experimental procedures and resulting kinetic parameters vary with the assumed kinetic model and applied data fitting procedure. In absence of a reliable model, reaction engineers are forced to scale-up the established reactors in economically undesirable small steps. Additionally, the available reaction data are insufficient for development of process and designing new reactor concepts with justifiable expenditure. These problems are addressed in this study.

In the present work, we have studied effects of reaction temperature and charge ratios as a function of time on batch reaction performance. Particle size and reaction time are optimized. Effect of time and PET particle size on acid value of TPA for PET depolymerization was studied. Simultaneous fragmentation and depolymerization model scheme was proposed. Reaction kinetic model was explained and experimental data were fitted to validate the model. EG was recovered by salting-out technique. Thermodynamic parameters (Ke, k, ΔG, ΔH and ΔS) required during reactor design or to scale-up the established reactor are determined.

EXPERIMENTAL

Materials, Chemicals and Reagents

PET waste used was procured from Garaware Polyesters, Aurangabad M.S. (INDIA). Here waste just mean materials left over after some products were made from raw material. Other materials used were neutral water, phenolphthalein, cyclohexylamine, HCl, methanol, ethanol, NaOH, KOH, Na_2SO_4, and dimethylsulfoxide (DMSO). Materials used were of analytical grade, supplied by s.d. fine chemicals (INDIA). Materials were used as received without further purification.

Alkaline Hydrolysis of PET

All alkaline hydrolysis reactions were carried out in a two-necked 1000 mL round bottom flask of borosil made as a batch reactor equipped with a reflux condenser, a thermometer and heating assembly with magnetic stirrer control arrangements, operated at atmospheric pressure.

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These reactions were carried out at 90–150°C at intervals of 20°C at atmospheric pressure. Reaction was carried out by taking 10 g PET and 5 g of Na_2SO_4 in 50 mL of various concentrations of NaOH with 4 mL of cyclohexylamine for different periods of time ranging from 10–110 min at the intervals of 10 min. Different particle sizes ranging from 50 to 512.5 μm of PET were taken for this reaction (separately). After conducting the reaction for the specified time, reaction mixture was filtered out and filtrate was boiled, and a stoichiometric amount of neutral water was added to dissolve disodium terephthalate (sodium salt of TPA) in it. Temperature of sodium salt of TPA solution was maintained at 80°C. Stoichiometric amount of HCl was added to convert sodium salt of TPA into TPA (during this step, addition of HCl in TPA salt resulted in the formation of precipitation of TPA). TPA was separated from rest of the materials, purified and dried until constant weight at 80°C in an oven. Weight of product (TPA) was measured.

EG salt was present in the resulting liquid phase that was converted into EG by using HCl. EG was concentrated and weighed.

Residual PET was washed with water, dried until constant weight at 80°C in an oven and weighed.

Percent depolymerization of PET, yield of TPA, and yield of EG were determined by gravimetry and defined in the following ways:

(i) Depolymerization of PET (%) $= \{(W_{PET,i} - W_{PET,R})/W_{PET,i}\} \times 100$.

(ii) Yield of TPA (%) $= \{m_{TPA,O}/m_{PET,i}\} \times 100$.

(iii) Yield of EG (%) $= (m_{EG,O}/m_{PET,i}) \times 100$.

where $W_{PET,i}$ is initial weight of PET, $W_{PET,R}$ is weight of unreacted PET, $m_{TPA,O}$ is number of moles TPA, $m_{EG,O}$ is number of moles of EG and $m_{PET,i}$, is initial number of moles of PET monomers.

Determination of Acid Value of TPA

Determination of acid value of TPA was carried out with 1 g of TPA powder by using 35 mL of dimethylsulphoxide (DMSO) in which phenolphthalein was added as an indicator. It was titrated against an ethanolic 0.1N KOH solution until a faint pink colour appeared (Dara, 1999; Vogel, 1939). Acid values were calculated with the relation:

$$\text{Acid value} = 5.611 \times (\text{volume of 0.1N KOH in mL})/(\text{weight of sample in g})$$

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Determination of Molecular Weight of Control and Residual PET

Viscosity average molecular weight (M) of control PET at 25°C was determined by Ostwald's viscosity method (Dara, 1999; Vogel, 1939). Solvent used was mixture of phenol and tetra-chloroethane in proportions of 3: 5 (v/v). Clean and dry Ostwald's viscometer was used for determination of time of flow for solvent mixture by keeping the viscometer in a thermostat in order to maintain constant temperature. Various solutions (0.001–0.006%) of control PET were prepared in solvent mixture. The flow times of each solution were also recorded. Effect of concentration of PET on viscosity is shown in Figure 1. The y-intercept of this plot was recorded as 27.7. Keeping the values of Mark–Houwink constants (Gowariker et al., 1990) ($k = 22.9 \times 10^{-3}$, $\propto = 0.73$) for solvent mixture and y-intercept = 27.7 (Figure 1), molecular weight was determined by the relation (Gowariker et al., 1990) $\eta_{sp}/C = k\ M^{\propto}$. Molecular weight determination of residual PET was also undertaken at 90, 110, 130 and 150°C by the same technique.

EG Recovery Method

For recovery of the monomeric product, EG from resulting liquid (contains mainly mixture of EG with excess water and unreacted NaOH) is

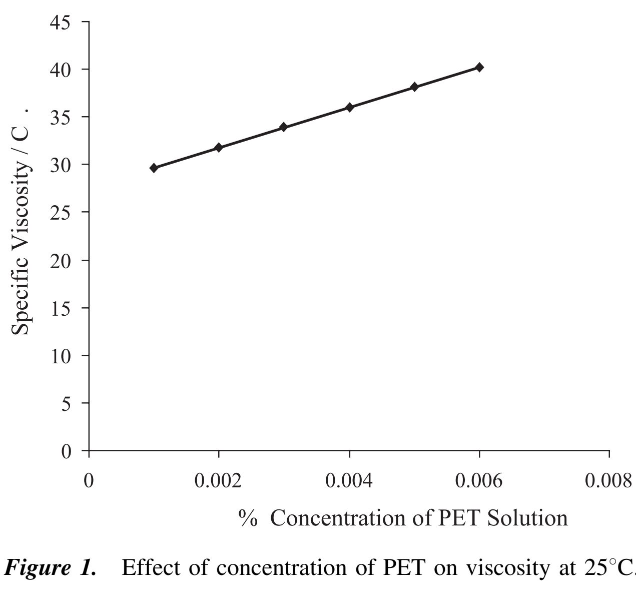

Figure 1. Effect of concentration of PET on viscosity at 25°C.

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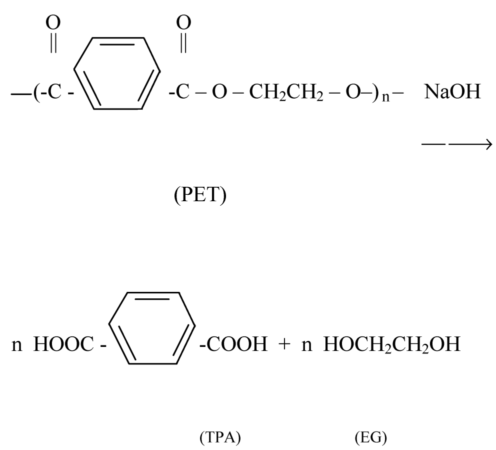
taken in a batch distillation assembly that is heated to 60°C. Sodium sulphate is introduced into the liquid mixture with constant stirring so as to obtain a saturated solution (Paszun and Spychaj, 1997). It was heated slowly at 125°C (below boiling point of EG 197°C, and above boiling point of water 100°C) until the excess water was almost distilled. As a result of this, EG salting-out takes place and a separate solid layer of disodium salt of EG (EG salt) is obtained. The batch distillation round bottom flask was cooled to room temperature (26°C) and the solid layer was separated from liquid phase (mainly alkali + H_2SO_4). EG was recovered from EG salt using stoichiometric amount of HCl that was analysed qualitatively and quantitatively. During the formation of EG salt the formation of H_2SO_4 takes place that is associated with unreacted NaOH. H_2SO_4 with unreacted NaOH was warmed for 5 min to convert it in the form of Na_2SO_4. This salting out method recovers complete monomeric EG.

Process Reaction of Depolymerization of PET in NaOH at 1 Atm



(1)

Chemical Reactions involved during EG Recovery Operations:

$$HOH_2C - CH_2OH + Na_2SO_4 \rightarrow NaOH_2C - CH_2ONa + H_2SO_4$$

$$NaOH_2C - CH_2ONa + 2HCl \rightarrow HOH_2C - CH_2OH + 2NaCl$$

$$H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2O$$

During separation of TPA, the following reactions take place:

$$\begin{aligned}&HOOC-C_6H_4-COOH+2NaOH\\&\quad\rightarrow NaOOC-C_6H_4-COONa+2H_2O\end{aligned}$$

$$\begin{aligned}&NaOOC-C_6H_4-COONa+2HCl\\&\quad\rightarrow HOOC-C_6H_4-COOH+2NaCl\end{aligned}$$

RESULTS AND DISCUSSION

Determination of Molecular Weight of Control and Residual Polyesters

Molecular weight of PET was recorded as 16703 at 25°C. Molecular weight determinations of residual polyester were also undertaken at 90, 110, 130 and 150°C (Table 1). Molecular weight of residual PET was slightly decreased with an increase of reaction temperature; however, for the case of highest temperature (i.e., 150°C), molecular weight of residual PET was 16641, still close to 16703. The slight decrease in molecular weight might be due to the decrease in viscosity with increase in temperature. Therefore, bulk phase of residual PET was close to fresh PET of long chains. This suggests that depolymerization of PET powder in NaOH solution occurred on external surface of the PET powder, and PET powder was lamellarly depolymerized.

Table 1. Average molecular weight of the residual PET after alkaline hydrolysis as a function of temperature. In each batch of the feed, the weight of PET is 10 g; charge ratio (M) is 4; reaction time is 90 min; total reactor volume is 1000 mL; PET particle size is 127.5 μm; at 1 atm.

Temperature (°C)	Average molecular weight of PET
25	16703
90	16674
110	16660
130	16648
150	16641

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Qualitative Analysis of Depolymerized Products of Waste PET

Liquid product obtained from depolymerization of PET was analyzed by a method suggested by Vogel (1939). Observed weight per mL (1.113 g), boiling point (195°C) and molecular weight (62), were close to value of EG.

Solid product obtained from depolymerization of PET was analyzed (Vogel, 1939) by potentiometer, showed concentration of solid product close to value of TPA, 10.8 m mol /g of TPA, sublimed at 300°C and the acid value of this solid product was recorded as 651. These results clearly indicate that TPA was the solid product obtained from depolymerization of PET.

Effect of Temperature and Charge Ratio on Yields of Product

Table 2 lists PET conversion and yields of TPA and EG for depolymerization in NaOH solution for 90 min at different temperatures (90, 110, 130 and 150°C) and different charge ratios (M = 1, 2 and 4). Charge ratio ($M = N_{Bi}/N_{Ai}$), where N_{Ai} is moles of PET repeating units charged in the

Table 2. Temperature and charge ratio effect on PET conversions, TPA and EG yields (%) for depolymerization of PET in NaOH solution. The weight of the PET is 10 g; reaction time is 90 min; total reactor volume is 1000 mL; PET particle size is 127.5 μm; at 1 atm.

Charge ratios (M)	Temperature (°C)	PET conv (%)	TPA yield (%)	EG yield (%)
1	90	21.27	21.16	21.22
1	110	22.1	22.01	22.06
1	130	23.27	23.15	23.24
1	150	24.63	24.52	24.58
2	90	42.54	42.45	42.51
2	110	44.13	44.04	44.1
2	130	46.56	46.48	46.54
2	150	49.27	49.16	49.25
4	90	85.12	85.02	85.09
4	110	88.24	88.16	88.22
4	130	93.13	93.05	93.11
4	150	98.52	98.46	98.5

batch reactor before the depolymerization reaction, and N_{Bi} is moles of NaOH charged in the batch reactor before the depolymerization reaction.

It is observed that PET conversion apparently increases with reaction temperatures and charge ratios. Yields of TPA and EG were almost equal to PET conversion. Results indicate that PET solid powder depolymerized in NaOH solutions at different charge ratios and at different temperatures produced monomers (TPA and EG) in the solution and this proceeded quantitatively according to chemical Eq. 1. However, difference in yields of TPA and EG was negligible because oxidation/carbonization of EG was avoided by addition of Na_2SO_4 during reaction.

Effect of Temperature, Charge Ratios, Particle Size and Reaction Time on Depolymerization of PET and Acid Values of TPA

Effect of Temperature

Figures 2 and 3 show results of effect of temperatures (90, 110, 130 and 150°C) at 1 atm for charge ratios (M) of 2 and 4 as a function of

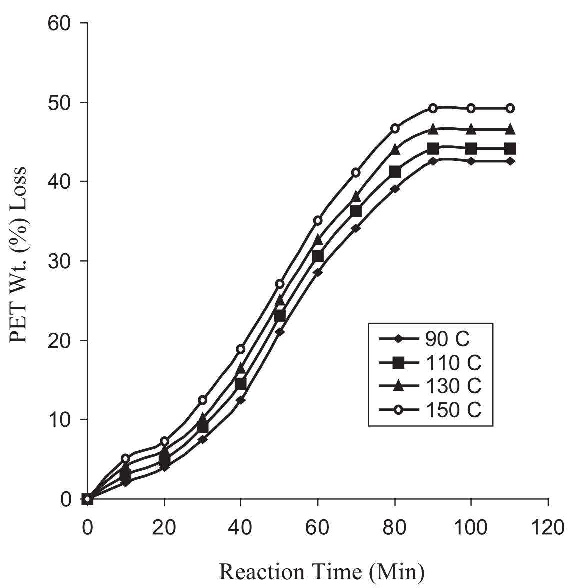

Figure 2. Effect of temperature (°C) on Wt (%) loss of 127.5 μm particle size of PET in NaOH (M = 2) at 90, 110, 130 and 150°C.

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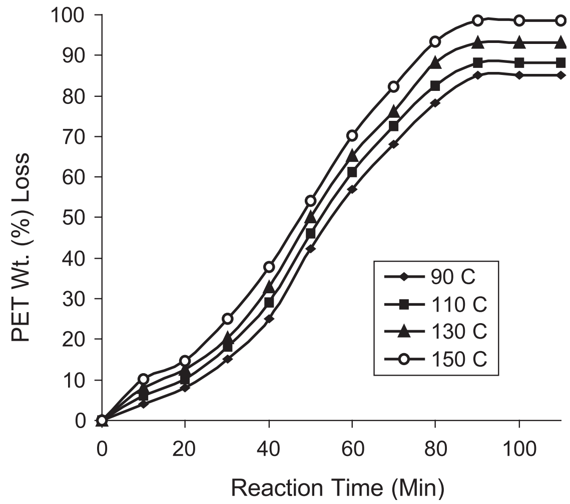

Figure 3. Effect of temperature (°C) on Wt (%) loss of 127.5 μm particle size of PET in NaOH (M = 4) at 90, 110, 130 and 150°C.

reaction time on PET wt. (%) loss during hydrolysis of PET, respectively. It was observed that on increasing temperature as well as charge ratios as a function of reflux time from 10 to 110 min, rate of wt (%) loss increased. Degree of hydrolysis was found proportional to reflux time.

Effect of Charge Ratios

Figure 4 shows results of charge ratios (M = 1, 2 and 4) as a function of reaction time at 150°C and 1 atm on the PET wt. (%) loss in NaOH solution for 127.5 μm particle size of PET. These results showed increment in PET wt. (%) loss with charge ratios as a function of reaction time. Degree of hydrolysis was found proportional to reflux time.

Optimization of Reaction Time

Optimal reaction time is the time at which the conversion of PET is maximum and thereafter with increase in time the conversion remains constant during alkaline hydrolysis of PET. From Figure 4 the maximum conversion of PET is 98.52 at 150°C using NaOH solution (M = 4) for the particle size of 127.5 μm. Maximum conversion of PET could be obtained by increasing the temperature upto 150°C easily rather than time,

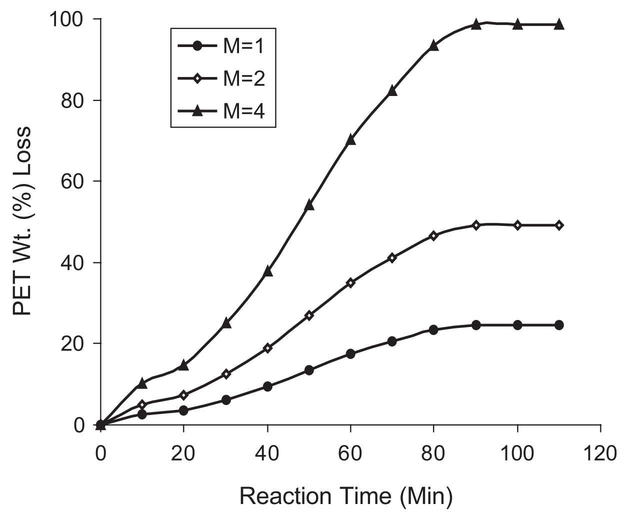

Figure 4. Effect of the charge ratios (M = 1, 2 and 4) on Wt (%) loss at 150°C for 127.5 μm particle size of PET.

and to save the reaction time. Hence, the optimal reaction time is recorded as 90 min.

Effect of Particle Size

Particle size for alkaline hydrolysis was optimized ranging from 50 to 512 μm at 90°C for 90 min (M = 4) (Figure 5). Wt. (%) loss of PET decreases with increase in particle size. Maximum and minimum wt. (%) loss for 90 min is observed for 50 and 512 μm, respectively. Optimal PET particle size is recorded as 127.5 μm.

Effects of particle size of PET at 90°C in NaOH (M = 4) for 90 min on acid value are illustrated in Figure 6. Acid values obtained for all depolymerized products are ranging from 651 to 489. Initially there is no change in acid value for 50–127.5 μm particle sizes. Acid value obtained for pure TPA is 652 of s. d. Fine Chemicals (657 of Sisco). Acid values are obtained for all depolymerized products (TPA). Acid values were determined at 98°C for 10–110 min time in NaOH (M = 4) for 127.5 μm PET particle. Acid values were significantly varied from 103 to 651 (Figure 7).

The result of acid value suggests that during fragmentation of PET powder initially there is formation of larger oligomers and TPA product in reaction mixture. Later larger oligomers are gradually converted into

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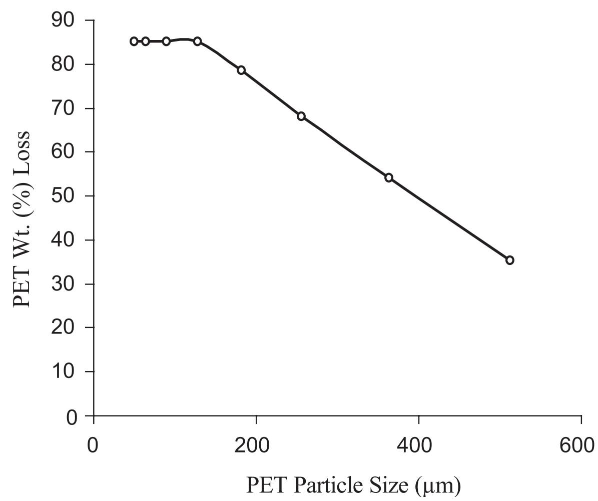

Figure 5. Optimization of PET size in alkaline hydrolysis and effect of particle size of PET on PET Wt. (%) loss at 90°C for 90 min reaction time for charge ratio (M) of 4 at 1 atm.

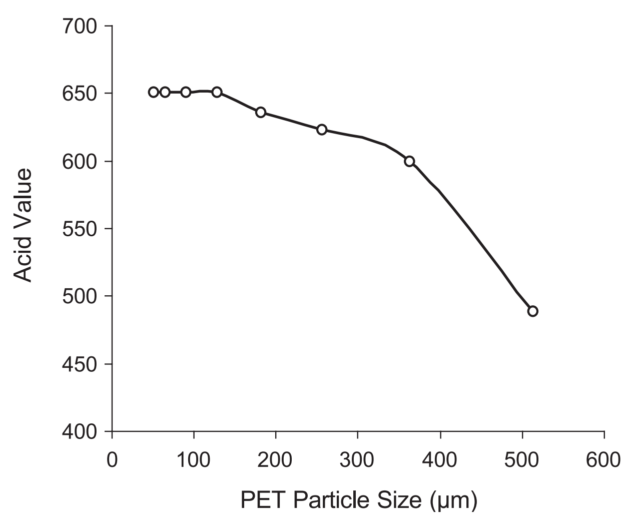

Figure 6. Effect of PET particle size on acid value of TPA at 90°C and 1 atm in NaOH (M = 4) for 90 min reaction time.

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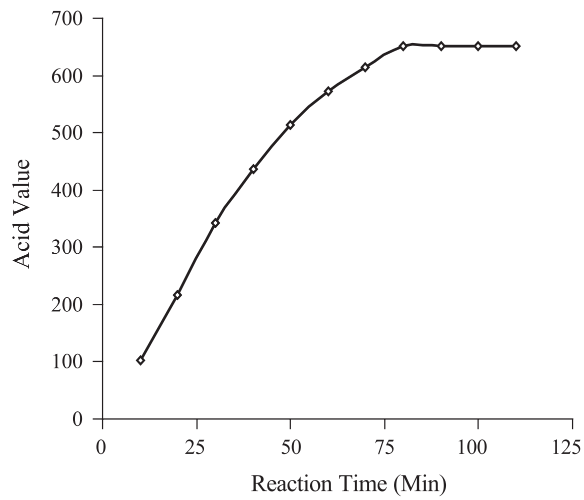

Figure 7. Effect of time on acid value of TPA at 90°C and 1 atm in NaOH (M = 4) for 127.5 μm PET particle size.

smaller oligomers and simultaneously TPA product is produced during depolymerization of PET as a function of reaction time. This supports to develop the model scheme for fragmentation and simultaneous depolymerization of PET (Figure 8).

EG Recovery

White precipitation test, colour stability test, and EG recovery results clearly indicate that EG does not undergo carbonization or oxidation at

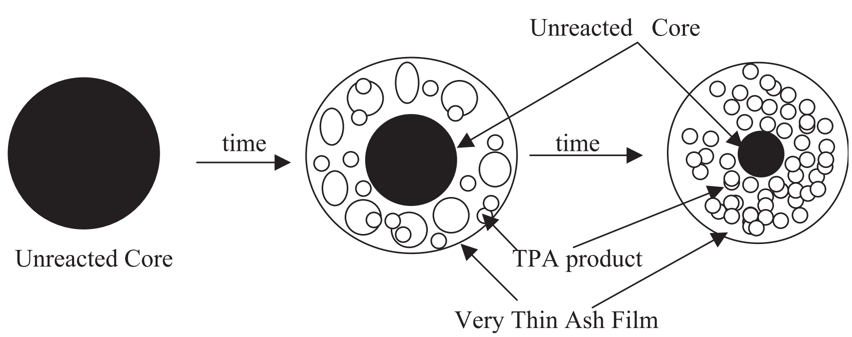

Figure 8. Simultaneous fragmentation and depolymerisation model scheme of PET particle in NaOH at 90°C for 90 min for charge ratio of 4 at 1 atm.

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Table 3. Effect of NaOH concentration on depolymerization of PET. The weight of PET is 10 g; reaction time is 90 min; solvent used is 40 mL; reactor volume is 1000 mL; PET particle size is 127.5 μm; at 1 atm and 90°C.

NaOH conc (N)	PET conv (%)
1	21.27
2	42.54
3	63.78
4	84.12

various reaction controlled parameters during depolymerization of PET in NaOH.

Kinetic Data

PET wt (%) loss at various temperatures and charge ratios (M) as a function of reflux time at 1 atm pressure was carried out (Figures 2–4). By increasing temperature and charge ratios as a function of reflux time, there was a significant increase in wt (%) loss of PET. Initial wt (%) loss was very low (1.01%) at 90°C (M = 1). Increasing with reaction time, difference of PET conversion in these cases increased. Reaction rate (slope of curve) was found to be almost proportional to charge ratios and temperatures as a

Table 4. Effect of PET concentration on PET conversion. Reaction time is 90 min; in 500 mL NaOH of 2.5 (mol/L) concentration; reactor volume is 1000 mL; PET particle size is 127.5 μm; at 1 atm and 90°C.

PET wt. (g)	PET wt. (%) loss
5	85
10	85
15	85
20	85
25	85
30	85

function of reflux time. Moreover, results (Table 3) clearly indicate that PET hydrolysis is proportional to NaOH concentration. Furthermore, Table 4 shows constant PET Wt. (%) loss as a function of PET concentration (cm^2/L). PET concentration (C_A) is the surface area of solid waste PET per unit batch reactor volume. Effect of PET concentration on PET Wt. (%) loss was examined at 90 °C for 90 min reflux time, while in experiments concentrations of NaOH were maintained constant (2.5 mol/L) and 5, 10, 15, 20, 25 and 30 g of PET were used in 500 mL of NaOH solution. It was found that same (85%) PET Wt. (%) loss was obtained for different charges of PET (Table 4). Furthermore, study of the effect of the PET concentration on the PET Wt. (%) loss as a function of time at 90°C and at 1 atm in NaOH of 2.5 mol/L concentration for 127.5 μm PET particle size was also undertaken. In the experiment various charges (10, 20, 30 and 40 g) of PET were introduced in 500 mL NaOH solution of 2.5 mol/L concentrations by varying the reflux time ranging from 20–90 min (Figure 9). Results (Figure 9) indicate that all the points (for the constant time) show the identical (i.e. constant) PET weight (%) loss for different charges (10, 20, 30 and 40 g) of PET. The PET Wt. (%) loss remains constant as a function of the PET concentrations, and it increases as a function of time

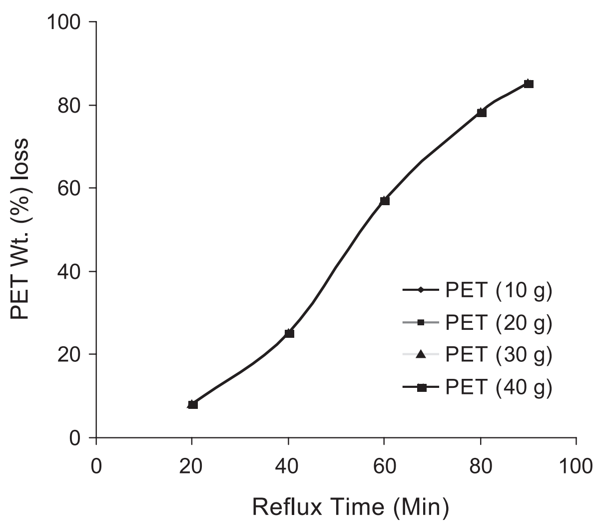

Figure 9. Effect of PET concentration on PET Wt (%) loss as a function of reaction time at 90°C and 1 atm in 500 mL of NaOH solution of 2.5 (mol/L) concentration; for 127.5 μm PET particle sizes.

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only (Figure 9). Figure 9 shows that the PET concentration would not affect the reaction rate and therefore the PET loss. These results clearly indicate that the initial reaction rates are exactly proportional to the concentrations of PET. Therefore, the reaction of the PET conversion in NaOH solution is first order to the PET concentration.

Based on the model scheme (Figure 8) the rate of hydrolysis is proportional to the surface area per unit volume of unreacted PET, and concentration of NaOH. In this model the effective surface area is proportional to the degree of unreacted PET (1-X), and there is an increase in conversion of PET as a function of reaction time. There occurs formation and growth of pores during depolymerization of a PET particle. Due to the growth (i.e. swelling) of PET particle the surface area of unreacted PET particle increases so that the PET loss increases as a function of reaction time.

Results (Table 4) indicate that the PET weight loss remains constant, which is the function of PET concentration but not the function of time. Hence, based on this model it would seem that the concentration of PET would neither affect the reaction rate nor the PET loss.

Kinetic Model

Results (Table 2) show that yields (%) of products (TPA and EG) were almost equal to PET conversion. Rate of reaction of alkali hydrolysis of PET can be defined by mole consumption rate of PET repeating unit per unit reactor volume. Product (TPA salt) produced in alkaline hydrolysis of PET was dissolved in solution. Disodium terephthalate (TPA salt) would be inactive to a nucleophilic substitution for esterification (reverse reaction for alkaline hydrolysis). Therefore, esterification will be negligible and overall reaction can be treated as irreversible (Ben-Zu et al., 2001). Hence, based on kinetic analysis, rate of hydrolysis of PET is given by:

$$-r_A = (N_{Ai}/V)(dX_A/dt) = k\, C_A C_B^b \tag{1}$$

where N_{Ai} represents moles of PET repeating units charged in solution, V represents reactor volume (L), X_A represents PET conversion, t represents reaction time (min), C_A represents concentration of PET (cm^2/L) that is the surface area of solid waste PET per unit batch reactor volume, C_B represents concentration of NaOH (mol/L), k is hydrolysis rate constant (L/min/cm^2), and b is reaction order with respect to NaOH. The surface of PET powder is available for reaction. Moreover, because PET particle size used is very small, so the shrinkage of particles will be much less compared to the original size for initial time, it can be neglected. Hence, a constant surface area of PET can be used during the course of initial reaction time periods (Ben-Zu et al., 2001). For a constant volume of a reaction vessel, the mass balance equation in terms of PET conversion (X_A) can be written as:

$$dX_A/dt = k\, N_{AS} V^{-b} N_{Ai}^{b-1} (M - 2X_A)^b \tag{2}$$

where N_{AS} represents surface area (cm^2) of PET powder per unit reactor volume (L), and N_{Bi} represents moles of NaOH charged in solution. From the model scheme as shown in Figure 8 the effective surface area of PET powder per unit reactor volume was treated as constant. M is the charge ratio defined as $M = N_{Bi}/N_{Ai}$. Based on experimental results, kinetic models in regard to rate order $b = 1$ and 2 are obtained as follows:

$$2k\, N_{AS} V^{-1} t = \ln[(M - 2X_{Ao})/(M - 2X_A)] \quad \text{for } b = 1 \tag{3}$$

$$2k\, N_{AS} V^{-2} N_{Ai} t = 1/(M - 2X_A) - 1/(M - 2X_{Ao}) \quad \text{for } b = 2 \tag{4}$$

$$\ln[(M - 2X_{Ao})/(M - 2X_A)] = Kt \quad \text{for } b = 1$$

$$1/[\ln 1 - (2X_A/M)] = Kt \quad \text{for } b = 1 \tag{5}$$

$$[1/(M - 2X_A)] - [1/(M - 2X_{Ao})] = Kt \quad \text{for } b = 2$$

$$X_A/[(M^2/2) - X_A M] = Kt \quad \text{for } b = 2 \tag{6}$$

where X_{Ao} represents PET conversion at zero reaction time and K represents apparent rate constant. These kinetic models were examined by fitting with experimental data. Three different charge ratios (M) were used at different temperatures at 1 atm. In case where the models fit reaction tendency well, relation of reaction time versus $1/[\ln 1\text{-}(2X_A/M)]$ from Eq. 5 and relation of reaction time versus $X_A/[(M^2/2)\text{-}X_A M]$ from Eq. 6 should be linear and slopes should be constant (independent of charge ratios, M). From fitting results, it is found that the model provides a good linear relationship with Eq. 5 for $b = 1$ (Figure 10); linear correlation factors (R^2) of fittings from Eq. 5 are above 0.9998 for different charge ratios and different temperatures. It was observed that Eq. 6 for $b = 2$ does not provide a linear relationship with data (Figure 11). Therefore, the model Eq. 6 for $b = 2$ is rejected. According to Eq. 5 plots/slopes from different charge ratios were exactly identical, however, plots/slopes from different temperatures were almost the same (Figure 10). In Figure 10 the value of R^2 for $b = 1$ with Eq. 5 is reported as 0.9998. This value is not 1. It shows that there is some error involved in the kinetic experiments or analysis. The existing negligible error might be due to shrinking of particle during hydrolysis. Moreover, negligible error might be due to some assumptions made during

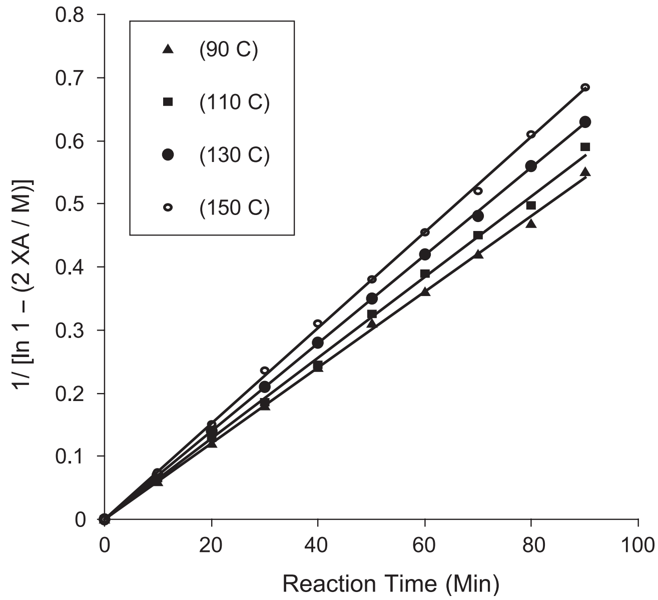

Figure 10. Effect of temperature as a function of time on rate of hydrolysis by model eqn 5 at 90, 110, 130 and 150°C in NaOH (M = 1, 2 and 4) and 1 atm for 127.5 μm PET particle size for fitting the reaction kinetic data.

modeling Eq. 5. The average value with a standard deviation of (1.97 + or − 0.28) × 10^{-3} was reported (Figure 10). Hence, Eq. 5, in which rate of reaction is close to first order to each PET concentration and first order to NaOH concentration, is a better model to describe kinetics of PET depolymerization in NaOH.

Thermodynamic Study of Hydrolysis of PET Using Compensation Effect

Arrhenius plot is shown in Figure 12. The activation energy (Ea) for alkaline hydrolysis of PET calculated from slope (at M = 4 and 90 min reaction time) is 26.3 kJ/mole, and respective Arrhenius constant (A) calculated from intercept (Figure 12) is 427.2 L/min/cm^2. Ea is also less than values from other researchers (Ben-Zu et al., 2001; Davies et al., 1962; McMahon et al., 1959; Ravens, 1960; Yoshioka et al., 1998, 2001) as shown in Table 5. Therefore, alkali hydrolysis is rate determining on PET surface. Variations in Ea and A values arise from a compensation effect.

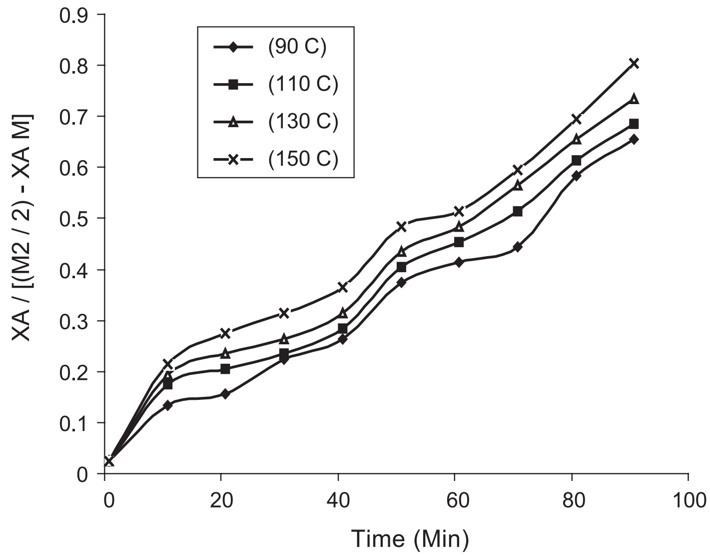

Figure 11. Effect of temperature as a function of time on rate of hydrolysis by model Eq. 6 at 90, 110, 130 and 150°C in NaOH (M = 1, 2 and 4) and 1 atm for 127.5 μm PET particle size for fitting the reaction kinetic data.

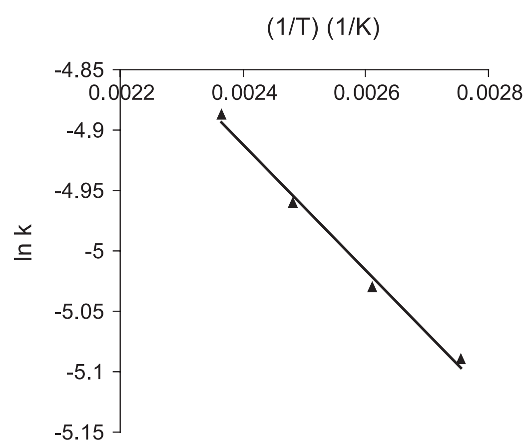

Figure 12. Arrhenius plot of the rate constant of hydrolysis in NaOH (M = 4) and 1 atm for 127.5 μm PET particle size.

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Table 5. Comparison of activation energy (Ea) and Arrhenius constant (A) as cited in literature for depolymerization of PET.

Ref. no.	Ea (KJ/mol)	A (L/min/cm^2)
Ours	26.3	427.2
Ben-Zu et al., 2001	69	419
Yoshioka et al., 1998	101.3	–
Yoshioka et al., 2001	88.7	–
McMahon et al., 1959 and Davies et al., 1962	109–113	–
Ravens, 1960	75.4–92.1	–

All Arrhenius plots will pass through a common point or very close to a point. Occurrence of this common temperature at which rate constants are identical leads to calling this relationship a theta rule. It seems reasonable to consider that a large number of sites of high energy will be formed at higher temperatures during preparation of the catalyst. These will be sites most active in catalysis since they will have various heats of adsorption and also energies of activation for surface reactions. So, high temperature preparations will have low energies of activation. Because there will be a few of these sites on low temperature preparations, energy of activation for reaction on these catalysts will be higher. But total number of sites will be greater and so a larger number of sites compensates for high energy of activation. A catalyst for which energy of activation is unfavorable will have a favorable number of active sites. Acceleration of a hydrolysis reaction rate caused by a hydrogen ion cannot be attributed to a change in Ea (or A) alone or rate constant. Ea is increasing when rate constant is decreasing and vice versa. Therefore, it is concluded that compensation effects occurred during alkaline hydrolysis of PET.

Table 6. Effect of temperature on hydrolysis rate constant (k), equilibrium constant (Ke), Gibbs free energy (ΔG), enthalpy (ΔH) and entropy (ΔS) of depolymerization of PET processed with NaOH solution (M = 4) for 1.5 h reaction time; for PET particle size 127.5 μm; at 1 atm pressure conditions.

Temp (K)	Ke	k (L/min/cm^2)	ΔG (kJ/kmol)	ΔH (kJ/kmol)	ΔS (kJ/kmol)
369	0.045	0.006156	9659	– 4921	– 39
383	0.049	0.006467	9603.5	– 5034	– 36
403	0.054	0.006967	9779.5	– 5063	– 33
423	0.058	0.007544	10013.5	– 5121	– 28

Hydrolysis rate constant, equilibrium constant, Gibbs free energy, enthalpy and entropy of hydrolysis of PET are determined (Table 6). The data mentioned in Table 6 were obtained as follows.

The values of k were determined by relation of k with K and Eq. 5.

$$1/[\ln 1 - (2X_A/M)] = Kt$$

Values of Ke were calculated by replacing C_{water} with C_{NaOH} in the Eq. 9 of (Campanelli et al., 1993). Values of the Gibbs free energy (ΔG) and enthalpy (ΔH) of hydrolysis of PETW were determined by Eqs. 13.11a and 13.14, respectively, of (Smith et al., 2001). The value of entropy (ΔS) was determined by the equation $(\Delta S) = (\Delta H - \Delta G)/T$, (p-478) of (Smith et al., 2001). Values of hydrolysis rate constant, equilibrium constant, Gibbs free energy, and entropy of reaction increase with temperature. But the value of the enthalpy of reaction decreases with increase in temperature. This shows that alkaline hydrolysis reaction of PET is endothermic and irreversible.

CONCLUSION

PET wt (%) loss increases almost linearly with increase in temperatures and charge ratios as a function of reaction time. Weight (%) loss of PET is inversely proportional to initial size of PET particles. Optimal PET particle size and reflux time are recorded as 127.5 μm and 90 min respectively. Yields (%) of TPA and EG are almost equal to PET conversion. Reaction kinetic model was explained and experimental data were fitted to validate the model, model shows a good agreement with data. EG is recovered by salting-out technique by introducing Na_2SO_4 during reaction. Based on results of acid values the model scheme of fragmentation and simultaneous depolymerization of PET particle is developed. Dependence of hydrolysis rate constant on reaction temperature was correlated by Arrhenius plot, which shows activation energy of 26.3 kJ/mol and Arrhenius constant of 427.2 L/min/cm^2.

NOTATION

$W_{PET,I}$	initial weight of PET
$W_{PET,R}$	weight of unreacted PET
$m_{TPA,O}$	number of moles of TPA
$m_{PET,i}$	initial number of moles of the PET monomer unit
V	reactor volume, L

k	rate constant per unit surface area, L/min/cm^2
N_{AS}	effective surface area for the reaction, cm^2
C_A	concentration of PET, surface area of solid waste PET per unit batch reactor volume, cm^2/L
X	degree of degradation of PET
K	apparent rate constant, equal to (3 k C_A/ro p)
Ke	equilibrium rate constant
ΔG	Gibbs free energy of reaction, (kJ/kmol)
ΔH	enthalpy of reaction, (kJ/kmol)
ΔS	entropy of reaction, (kJ/kmol)

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